

Nadine Berner

Lead Data Scientist



CONTACT

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CORE SKILLS

- Time Series Forecasting (Anomaly Detection, NeuralProphet, Deep Learning)
- Geospatial Analytics (Spatial Statistics, Mobility Patterns, Network Graphs)
- Machine Learning (Adaptive Sampling, Dimensional Reduction)
- Uncertainty Quantification (MCDET, PRA, PSA)
- Python, SQL, MLflow, AWS

AI & AGENTS

- LLM Agents & Tool Use
- Time Series Foundation Models (Chronos)
- GitHub Copilot (Custom Instructions & Tools)
- LLM-assisted Analytics
- LangChain

LANGUAGES

- German (native)
- English (fluent)
- French (elementary)
- Japanese (elementary)

Profile

Lead Data Scientist with over 10 years across time series forecasting, geospatial data science, & machine learning — grounded in a PhD in theoretical physics applying Bayesian inference to complex climate time series. Delivered production systems at BMW Group spanning anomaly detection, electric vehicle fleet forecasting for energy trading, & geospatial data quality for autonomous driving functions, building on earlier research & tool design in probabilistic safety analytics. For the past two years, extended this into AI systems & agentic approaches, building LLM-based multi-agent applications.

Professional Experience

Lead Data Scientist | Location-based Intelligence — BMW Group AG

2023 – present | Munich

- Introduced anomaly detection into the worldwide map-data release process combining an ensemble of statistical methods to generate anomaly scores & patterns, synthesized into executive reports via LLM
- Built forecasting capability for electric vehicle fleet charging behavior using NeuralProphet to power a prediction API that enables to participate in intra-day energy trading
- Designed a multi-agent system (LangChain) that resolves free-text customer queries into actionable next-destination suggestions, using a points-of-interest database via a REST API; validated feasibility & result quality as an MVP, & fed insights back to the database team to optimize data structures for LLM-based usage
- Led a time series forecasting use case at a BMW Group × AWS hackathon, evaluating the Chronos time series foundation model; direct exchange with the paper's authors sharpened the team's understanding of the capabilities & limitations of time series LLMs

Senior Data Scientist | Data Quality & Algorithms — BMW Group AG

2022 – 2023 | Munich

- Designed a data lake filled by a worldwide map-data quality pipeline on AWS (S3, DynamoDB, Athena) — including sandboxes (SageMaker) for in-depth spatial analyses supporting autonomous driving functions; for China, co-developed a cloud-agnostic approach with a dedicated dev team to meet local infrastructure requirements
- Implemented automated quality monitoring via dashboards (Plotly/Dash, Kibana), accross markets & map products
- Advised as lead data scientist on suitable methods, partnering with Tech Offices in Asia to incorporate on-the-ground data expertise into the analyses

Data & Software Architect | Prototype Integration — BMW Group AG

2021 – 2022 | Munich

- Translated technical system knowledge of vehicle E/E systems into actionable guidance for prototype integration in the plant

Research Data Scientist | Safety Analysis — Gesellschaft für Anlagen- und Reaktorsicherheit (GRS) gGmbH

2014 – 2021 | Garching

- Developed Monte Carlo Dynamic Event Tree method for probabilistic safety analysis of stochastic, complex dynamic systems, in Python — orchestrating legacy Fortran/C++ simulation code via a custom wrapper mechanism
- Developed adaptive sampling with Gaussian processes & ML to localize cliff-edge effects in complex system dynamics
- Designed multi-hazard dependency network analysis for visualizing and quantifying cascading impacts in plant systems
- Published in public research reports & shared as advanced analytical toolboxes for researchers & safety experts

Education

Dr. rer. nat., Theoretical Physics

University of Potsdam, Germany, 2016

Thesis: Deciphering Multiple Changes in Complex Climate Time Series using Bayesian Inference

Full list of publications, teaching & mentoring: nadineberner.eu